Cambridge O Level Computer Science

Chapter 6: Automated and emerging technologies

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- Automated systems: the use of sensors, microprocessors and actuators, and the advantages and disadvantages of automation in industry, transport, agriculture, weather, gaming, lighting and science.
- Robotics: what robotics means, the characteristics of a robot, independent vs dependent robots, and the roles, advantages and disadvantages of robots.
- Artificial intelligence: what AI is, its main characteristics, expert systems, and machine learning.

From robots welding cars to autonomous lawn mowers and smart road signs, automated systems are everywhere and growing more sophisticated. This chapter looks at how they work and how they affect everyday life. Note: in exam questions, any industrial or scientific process will be fully described to you (possibly with a diagram); what you must be able to do is explain the interaction between the sensors, actuators and microprocessor/computer.

6.1 Automated systems

6.1.1 Sensors, microprocessors and actuators

An automated system is a combination of software and hardware (sensors, microprocessors and actuators) designed and programmed to work automatically without human intervention, though human monitoring is often involved. The pattern is always the same (see Chapter 3):

- Sensors are input devices that take readings from their surroundings and send the data to a microprocessor or computer; analogue data is first converted to digital by an ADC.
- The microprocessor processes the data and decides what action to take, based on its programming.
- The output is usually signals sent to actuators, which control motors, wheels, solenoids, valves, pumps and so on.

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For every example in this section, ask: which sensors are used? What do the actuators do? What does the computer do?

Check yourself

1. 1. What is an automated system?
2. 2. Describe the three-stage pattern of sensor, microprocessor and actuator in an automated system.
3. 3. Why is an ADC needed between sensors and the microprocessor?

6.1.2 Automated systems in industry

Example 1: A nuclear power station

At the centre is a distributed control system (DCS): a powerful computer programmed to monitor and control the whole process with no human interaction. Sensors (temperature, pressure, flow level, escaping gas, radiation level) send data to the DCS, consulting a large database of operational parameters; the DCS then signals actuators to operate water pumps, valves, gas pumps and even the automatic shutdown system. A human supervisor in a remote control room watches a schematic on a large screen and can override the DCS and shut the process down.

Example 2: Manufacture of paracetamol

Two processes run side by side: Process 1 makes the paracetamol, Process 2 forms the solid tablets. Sensors in both send data to a central computer, which checks a database to confirm both processes are within correct parameters, then signals actuators to operate pumps, valves, heaters, stirrers and pistons. A remote monitoring station with a human operator can override the system if necessary.

Advantages of these industrial automated systems: much faster reactions than a human operator; safer (timely interventions, and humans kept away from dangerous environments); processes run under optimum conditions; less expensive in the long run (no 24-hour workforce); more efficient use of materials; higher productivity; more consistent results. Disadvantages: expensive to set up and needs extensive testing; unforeseen conditions may never have been tested (hence the supervisor); vulnerable to cyberattacks; enhanced (expensive) maintenance needed.

Check yourself

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4. 1. For the nuclear power station, state (a) three types of sensor used, (b) what the DCS does, (c) two things the actuators operate.
5. 2. Why does the nuclear power station still have a human supervisor even though the DCS is fully automatic?
6. 3. Give three advantages and three disadvantages of automating the paracetamol manufacturing process.
7. 4. In car engine manufacture, manual piston installation had an error rate of about 1.5%, while automated systems reduced it to 0.00001%. What does this tell you about automation and consistency?

6.1.3 Automated systems in transport

Example 3: Self-parking cars

The driver drives along the row of parked cars while on-board sensors and cameras gauge the size of parking spaces; the computer warns the driver when a suitable space is found. The driver selects auto-parking and the computer takes over completely: actuators operate the steering rack, brakes and throttle. Sensors in the bumpers both transmit signals and receive the reflections; the time for the signal to return lets the computer calculate object positions and build a 3D image of the surroundings, so the car fits into the space automatically. (Cheaper, older systems only control the steering; the driver works the brakes and throttle.)

Advantages: fewer parking spaces needed for the same number of cars; less traffic disruption in cities; cars fit into smaller spaces; fewer dents and scratches (fewer insurance claims); safer, since sensors watch for objects including small children (the manoeuvre stops if a new object appears); very consistent results.
Disadvantages: drivers become over-reliant and lose skills; faulty or dirty sensors/cameras can send false data and cause malfunctions; sensors may miss low kerbs (kerbing of wheels); an expensive option that saves the driver no money; extra maintenance required.

Example 4: Adaptive cruise control

The driver sets a cruising speed (for example, 100 kph) on a touch screen. Lasers in the bumpers constantly send out signals that bounce off the vehicle in front and return to the car's sensors; the return time lets the on-board computer calculate the distance between the vehicles. If the distance is less than the safe distance, the computer signals actuators to apply the brakes and/or reduce the throttle. If the

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distance is safe, the computer checks whether the current speed matches the set speed and signals the actuators to increase or decrease the throttle accordingly.

Worked example: A car with adaptive cruise control is set to 100 kph. The book's table gives a stopping distance of 77 m at 100 kph. The sensors measure the distance to the vehicle in front as 60 m. Explain what the computer does. Then consider a second case: the car is set to 60 kph (stopping distance 32 m) and the measured distance is 50 m with the car already travelling at 60 kph.

Case 1: 60 m < 77 m, so the distance is less than the safe distance. The computer sends signals to the actuators to apply the brakes and reduce the throttle until a safe gap is restored.

Case 2: 50 m > 32 m, so the distance is safe. The computer then checks the speed: it already equals the set value of 60 kph, so it signals the actuators to maintain the throttle (no change needed).

Check yourself

8. 1. Describe how bumper sensors help a self-parking car build a 3D image of its surroundings.
9. 2. Give three advantages and three disadvantages of self-parking cars.
10. 3. Explain how adaptive cruise control uses lasers to keep a safe distance from the vehicle in front.
11. 4. A car is set to 80 kph (stopping distance 53 m) and the sensors measure 70 m to the vehicle ahead. What will the computer do? Explain your answer.

6.1.4 Automated systems in agriculture and weather

Example 5: Automated irrigation in Brazil

Used on vast fields (some over 10 km2). An automatic weather station sends data to the controller every ten minutes (vital if very wet or very dry conditions are predicted). Ultrasonic water-level sensors in the irrigation channels send data back via wireless transmitters to a wireless receiver and then to the controller. The controller combines both data sources and decides whether to start or stop a series of water pumps, signalling actuators to operate them. A supervisor monitors

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